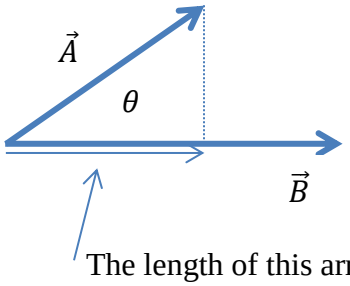
Physics Department
Camosun College
DOT PRODUCT and CROSS PRODUCT of VECTORS

PHYSICS 191/154 Tutorial

DOT PRODUCT:

$\vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|cos\theta$, where θ is the angle between vectors $\vec{A}$ and $\vec{B}$
The result is a SCALAR Quantity. The order of operation makes no difference: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$

A VISUALIZATION of the DOT PRODUCT is the PROJECTION of a VECTOR onto another VECTOR. Here, the PROJECTION of VECTOR $\vec{A}$ onto VECTOR $\vec{B}$:



The length of this arrow ($Acos\theta$) shows the fraction of vector $\vec{A}$ that points along the direction of vector $\vec{B}$ (the projection of $\vec{A}$ onto $\vec{B}$).

The DOT PRODUCT of any vector with itself is the square of the magnitude of the vector:

$\vec{A} \cdot \vec{A} = |\vec{A}||\vec{A}|cos\theta = |\vec{A}|^2$ since $\theta = 0^o$ and $\cos(0^o) = 1$

DOT PRODUCT of UNIT VECTORS:

$\vec{\imath} \cdot \vec{\jmath} = |\vec{\imath}||\vec{\jmath}|cos\theta = 1 \cdot 1cos(90^o) = \vec{\jmath} \cdot \vec{\imath} = |\vec{\jmath}||\vec{\imath}|cos\theta = 1 \cdot 1cos(90^o) = 0$
$\vec{\imath} \cdot \vec{\imath} = |\vec{\imath}||\vec{\imath}|cos\theta = 1 \cdot 1cos(0^o) = \vec{\jmath} \cdot \vec{\jmath} = |\vec{\jmath}||\vec{\jmath}|cos\theta = 1 \cdot 1cos(0^o) = 1$

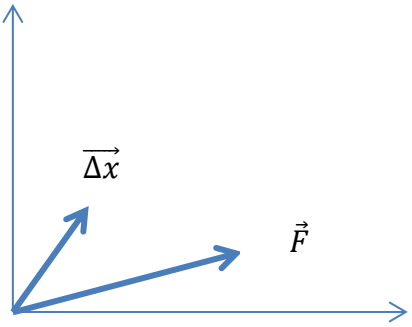
DOT PRODUCT of VECTORS in Unit Vector Notation:

$\vec{A} \cdot \vec{B} = (a_x\vec{\imath} + a_y\vec{\jmath}) \cdot (b_x\vec{\imath} + b_y\vec{\jmath}) =$
$= \left((a_xb_x)\vec{\imath} \cdot \vec{\imath} + (a_xb_y)\vec{\imath} \cdot \vec{\jmath} + (a_yb_x)\vec{\jmath} \cdot \vec{\imath} + (a_yb_y)\vec{\jmath} \cdot \vec{\jmath}\right) = (a_xb_x) + (a_yb_y)$

Simplifying…

$\vec{A} \cdot \vec{B} = (a_xb_x) + (a_yb_y)$ ← easy to remember!

DOT PRODUCT is used to calculate Mechanical WORK: W = $\vec{F} \cdot \overrightarrow{\Delta x} = \left|\vec{F}\right|\left|\overrightarrow{\Delta x}\right| cos\theta$



Example 1: A force of $\vec{F} = 5.0\ N\ @\ 10^o$ acts on an object that moves through a displacement of $\overrightarrow{\Delta x} = 2.5\ m\ @\ 55^o$. How much Work was done?

Answer: The angle between $\vec{F}$ and $\overrightarrow{\Delta x}$ is 45°.

So W = $\vec{F} \cdot \overrightarrow{\Delta x} = \left|\vec{F}\right|\left|\overrightarrow{\Delta x}\right| cos\theta = (5.0N)(2.5m)cos(45^o) = 12.5\ J\ cos(45^o) \cong 8.8\ J$, rounded to 2 sig fig.

Example 2: Repeat Example 1 with vectors converted to $\vec{\imath}\vec{\jmath}$ notation.

Answer:

$\vec{F} = 5.0\ N\ @\ 10^o \cong (4.92\vec{\imath} + 0.868\vec{\jmath})N$ and $\overrightarrow{\Delta x} = 2.5\ m\ @\ 55^o \cong (1.43\vec{\imath} + 2.05\vec{\jmath})m$.

So $\vec{F} \cdot \overrightarrow{\Delta x} = (4.92\vec{\imath} + 0.868\vec{\jmath})N \cdot (1.43\vec{\imath} + 2.05\vec{\jmath})m = \{(4.92 \times 1.43) + (0.868 \times 2.05)\}J =$
$= \{7.03 + 1.78\}\ J \cong 8.8\ J$, rounded to 2 sig fig. Same as Example 1.

Practise Problems (Calculate the Work done in each case):

1. Force $\vec{F} = 15N\ @\ 10^o$, displacement $\overrightarrow{\Delta x} = 115\ cm\ @\ 0^o$ [ans 17 J]
2. Force $\vec{F} = (2.0\vec{\imath} + 8.0\vec{\jmath})N$, displacement $\overrightarrow{\Delta x} = (-6.2\vec{\imath} - 4.5\vec{\jmath})cm$ [ans -0.48 J approx]
3. Force $\vec{F} = 2500N\ @\ 30^o\ E\ of\ N$, displacement $\overrightarrow{\Delta x} = 60.0m\ @\ 60^o\ N\ of\ W$ [ans 75 kJ]
4. Force $\vec{F} = (-150\vec{\imath} - 450\vec{\jmath})N$, displacement $\overrightarrow{\Delta x} = (3.6\vec{\imath} + 1.2\vec{\jmath})m$ [ans -1.1 kJ approx.]
5. Find the angle between the $\vec{F}$ vector and the $\overrightarrow{\Delta x}$ vector in problem 2. and 4. (above).

   [ans. 2.: 94° approx.; 4.: 127° approx.]

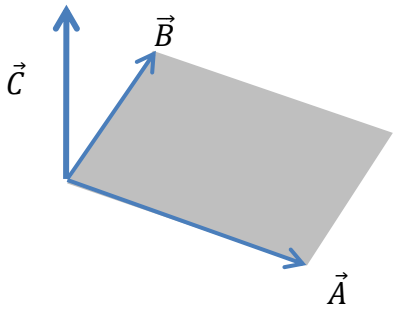
CROSS PRODUCT:

Definition:

$\vec{A} \times \vec{B} = \vec{C} \quad |\vec{C}| = |\vec{A} \times \vec{B}| = |\vec{A}||\vec{B}|sin\theta$, where θ is the angle between vectors $\vec{A}$ and $\vec{B}$

The result is another VECTOR Quantity.
The order of operation is important! $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$

A VISUALIZATION of the CROSS PRODUCT is a 2D plane defined by vectors $\vec{A}$ and $\vec{B}$, with the CROSS PRODUCT vector $\vec{C}$ perpendicular to that plane:



The direction of $\vec{C}$ is determined by the "Right Hand Rule": Vector $\vec{C}$ points in the direction that a Right-Hand screw thread would advance if rotated in the same sense as rotating vector $\vec{A}$ into or towards vector $\vec{B}$.

The Resultant Vector $\vec{C}$ is perpendicular to the 2D plane defined by vectors $\vec{A}$ and $\vec{B}$.

The CROSS PRODUCT of any vector with itself is ZERO.

$\vec{A} \times \vec{A} = \vec{C} \quad |\vec{C}| = |\vec{A} \times \vec{A}| = |\vec{A}||\vec{A}|sin(0^o) = 0,$

The CROSS PRODUCT of UNIT VECTORS:

$\hat{\imath} \times \hat{\imath} = \hat{\jmath} \times \hat{\jmath} = \hat{k} \times \hat{k} = 0$
$\hat{\imath} \times \hat{\jmath} = \hat{k} \quad \hat{\jmath} \times \hat{k} = \hat{\imath} \quad \hat{k} \times \hat{\imath} = \hat{\jmath}$

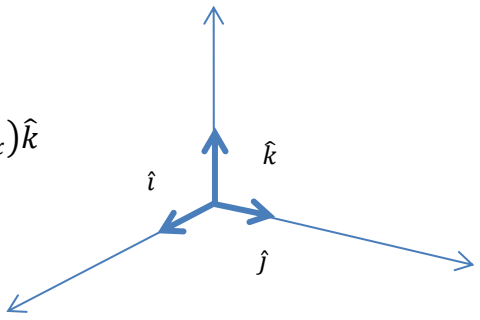
The CROSS PRODUCT of VECTORS in Unit Vector Notation:

Special Case of vectors $\vec{A}$ and $\vec{B}$ in the XY plane:

$\vec{A} \times \vec{B} = (a_x\hat{\imath} + a_y\hat{\jmath}) \times (b_x\hat{\imath} + b_y\hat{\jmath}) = (a_xb_y - a_yb_x)\hat{k}$
NOTE: the answer is a vector perpendicular to the XY plane

General Case of vectors $\vec{A}$ and $\vec{B}$ in any direction in 3D:

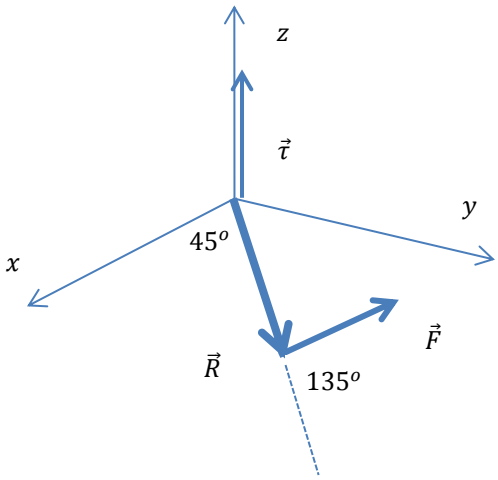
$\vec{A} \times \vec{B} = (a_x\hat{\imath} + a_y\hat{\jmath} + a_z\hat{k}) \times (b_x\hat{\imath} + b_y\hat{\jmath} + b_z\hat{k}) =$
$= (a_yb_z - a_zb_y)\hat{\imath} - (a_xb_z - a_zb_x)\hat{\jmath} + (a_xb_y - a_yb_x)\hat{k}$



CROSS PRODUCT is used to calculate TORQUE:

Torque is a Vector Quantity (has directions and magnitude).

$\vec{\tau} = \vec{R} \times \vec{F}$ $\quad |\vec{\tau}| = |\vec{R}||\vec{F}|sin\theta$ (this is the familiar equation for the magnitude of a torque)

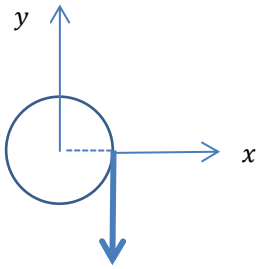


Example: A force of $\vec{F} = 5.0N$ @ 180° in the XY plane acts on a lever arm $\vec{R} = 2.5\ m$ @ 45° also in the XY plane.

What is the torque on the lever?

Answer:
The magnitude of the torque is $|\vec{\tau}| = |\vec{R}||\vec{F}|sin\theta = (2.5m)(5.0N)sin(135^o) \approx 8.84\ Nm$
[Note: the units are NOT Joules!]

The direction of the torque vector is found by rotating vector $\vec{R}$ into (towards) vector $\vec{F}$. A right-hand thread rotated in the same direction would advance in the **positive z-direction**. This is the direction of the torque vector, $\vec{\tau}$.



Practise Problems (Calculate the torque in each case):

1. Force $\vec{F} = 10\ N\ @ = -90^o$ acts on the outer edge of a disk with diameter 28.0 cm.
   [ans. $\vec{\tau} = (-1.4\ Nm)\hat{k}$, a direction *into* the paper, which would be in the negative z-direction; the Lever is indicated by the dashed line, and the Force indicated by the heavy arrow]

2. Force $\vec{F} = 23.5\ N\ @\ in\ an\ Easterly\ direction\ 45^o\ above\ the\ horizontal$ acts on a Lever $\vec{R} = 1.80m\ @\ East.$
   [ans. $29.9\ Nm\ @\ South$]

3. Force $\vec{F} = (2.0\vec{\imath} + 8.0\vec{\jmath})N$ acts on a Lever $\vec{R} = (-6.2\vec{\imath} - 4.5\vec{\jmath})cm.$
   [ans. $-0.406\ Nm\ \hat{k}$]

4. Force $\vec{F} = 8.25\ N\ @\ 76^o$ acts on a Lever $\vec{R} = 7.66\ cm\ @\ 216^o$. Both vectors are in the XY plane. This is the same as problem 3. (above) with the Force and Lever vectors expressed in Polar Notation.
   [ans. $-0.406\ Nm\ in\ the\ negative\ z-direction$]